

Strain-Engineered Oxygen-Modified Nickel Telluride/Nickel Oxide Heterostructures for Bifunctional Alkaline Water Electrocatalysis

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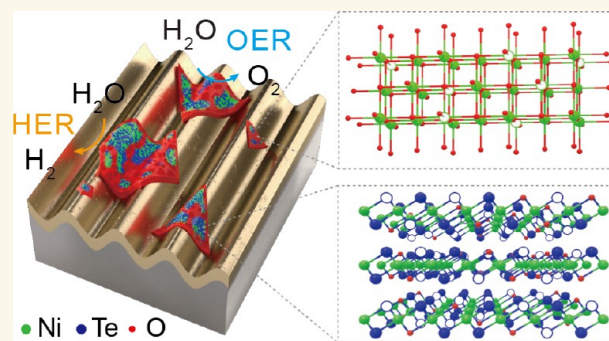
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ABSTRACT: We present a strain-engineering strategy for oxygen-modified nickel telluride/nickel oxide heterostructures capable of enabling bifunctional alkaline water electrolysis with performance surpassing Pt and IrO_x benchmarks. The heterostructures are synthesized via electrochemical Te dissolution and mild oxidation of mechanically exfoliated NiTe₂, followed by controlled strain induction through substrate buckling. Atomic-scale simulations and spectroscopic analyses indicate that Te-vacancy/O-substituted NiTe₂ domains promote oxygen-intermediate spillover between adjacent active sites, reducing OER overpotentials. In parallel, strained NiTe₂ domains facilitate hydrogen-intermediate transfer to NiO containing Ni vacancies, leading to accelerated HER kinetics and near-thermoneutral hydrogen adsorption. Strain modulation adjusts the electronic structure and increases active-site density, enabling stable operation at industrial-level current densities (>1 A cm⁻²). These findings illustrate how defect chemistry coupled with strain engineering can be utilized to develop high-performance, earth-abundant bifunctional electrocatalysts.

KEYWORDS: water electrolysis, hydrogen evolution reaction, oxygen evolution reaction, catalytic materials, transition metal dichalcogenides, strain engineering, electrochemical process



INTRODUCTION

Water electrolysis, including both the cathodic hydrogen evolution reaction (HER) and the anodic oxygen evolution reaction (OER), is important for producing molecular hydrogen and hydrogen fuels powered by renewable electricity without the emission of carbon dioxide.^{1–5} The main approach for industrial-level HER/OER is electrolysis of water in an alkaline solution in which the stability of constituent electrode materials is able to be preserved.^{6–9} The essential electrocatalysts, Pt-based and Ir/Ru-based materials for the alkaline HER and OER, respectively, however, have limited the scalability of water splitting because these noble-metal elements significantly decrease the cost-effectiveness of system-level electrolyzers while also generating supply chain risks.^{10–15} Although bifunctional catalysts for both HER and OER simplify the design of electrochemical devices and enable practical hydrogen energy systems, most catalysts have been optimized for either HER or OER because integrating bifunctional catalytic sites into a single material is challenging.^{12,16,17} Hence, the development of bifunctional electro-

catalysts based on readily available materials capable of enhancing the efficiency (and stability) of alkaline HER and OER simultaneously is crucial.

Transition metal dichalcogenides (TMDs) such as MoS₂ and WS₂ have exhibited promise as noble-metal-free electrocatalysts for HER because of their low cost and high per-site HER activity.^{2,18–23} In particular, strain-engineering of layered MoS₂ and MoTe₂ on viscoelastic substrates followed by electrochemical activation has enabled an acidic HER comparable to pure Pt.^{21,22,24,25} The HER on these materials system in an alkaline electrolyte, however, is problematic due to the sluggish water dissociation kinetics, which hampers the initial Volmer step in HER.^{26,27} The formation of Ni, Co, Fe,

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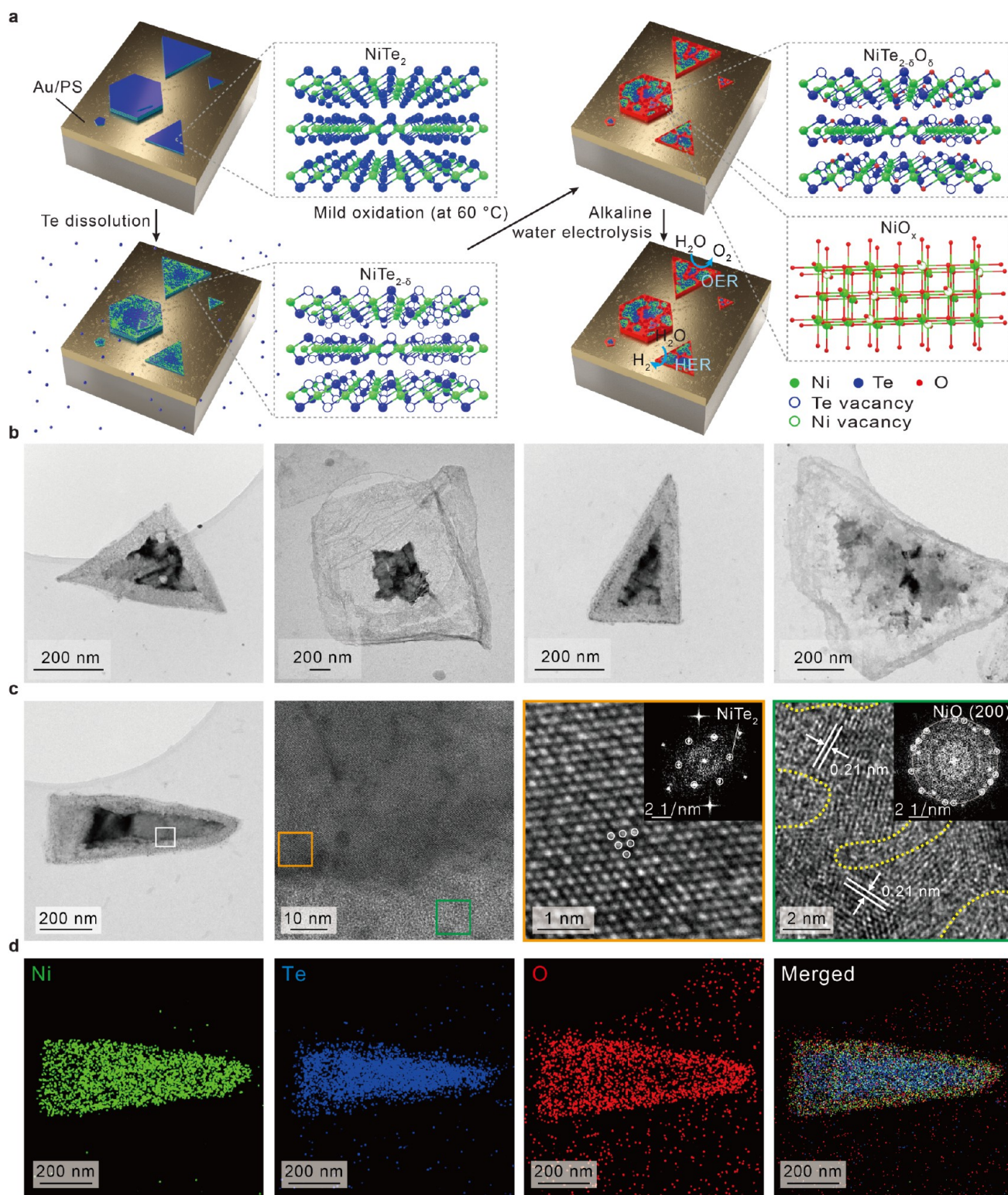


Figure 1. Forming procedure and TEM analyses of $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ heterostructures for alkaline water electrolysis. (a) Schematic illustration of electrochemical Te dissolution from mechanically exfoliated NiTe_2 flakes followed by oxidation to form heterostructured electrocatalysts on Au/PS substrates. (b–c) HRTEM images of heterostructures with different lateral lengths verifying $\text{NiTe}_{2-\delta}\text{O}_\delta$ and NiO_x domains with distinct crystalline and amorphous structures. Local areas enclosed with yellow dotted lines indicate the amorphous phase of NiO_x . (d) EDX compositional maps of the $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ heterostructure in (c).

and Mn hydroxides on MoS_2 and WS_2 has improved the alkaline HER by virtue of effective dissociation of water molecules with reduced barrier energies at heterointerfaces.^{28,29} Heterostructures consisting of $\text{Ni}(\text{OH})_2$ nanoparticles

and MoS_2 nanosheets have achieved the alkaline HER and OER simultaneously; the electrocatalysts also reduced the formation energies of oxygen intermediate species in the OER (i.e., $^*\text{OH}$, $^*\text{O}$, and $^*\text{OOH}$) during the four-electron transfer

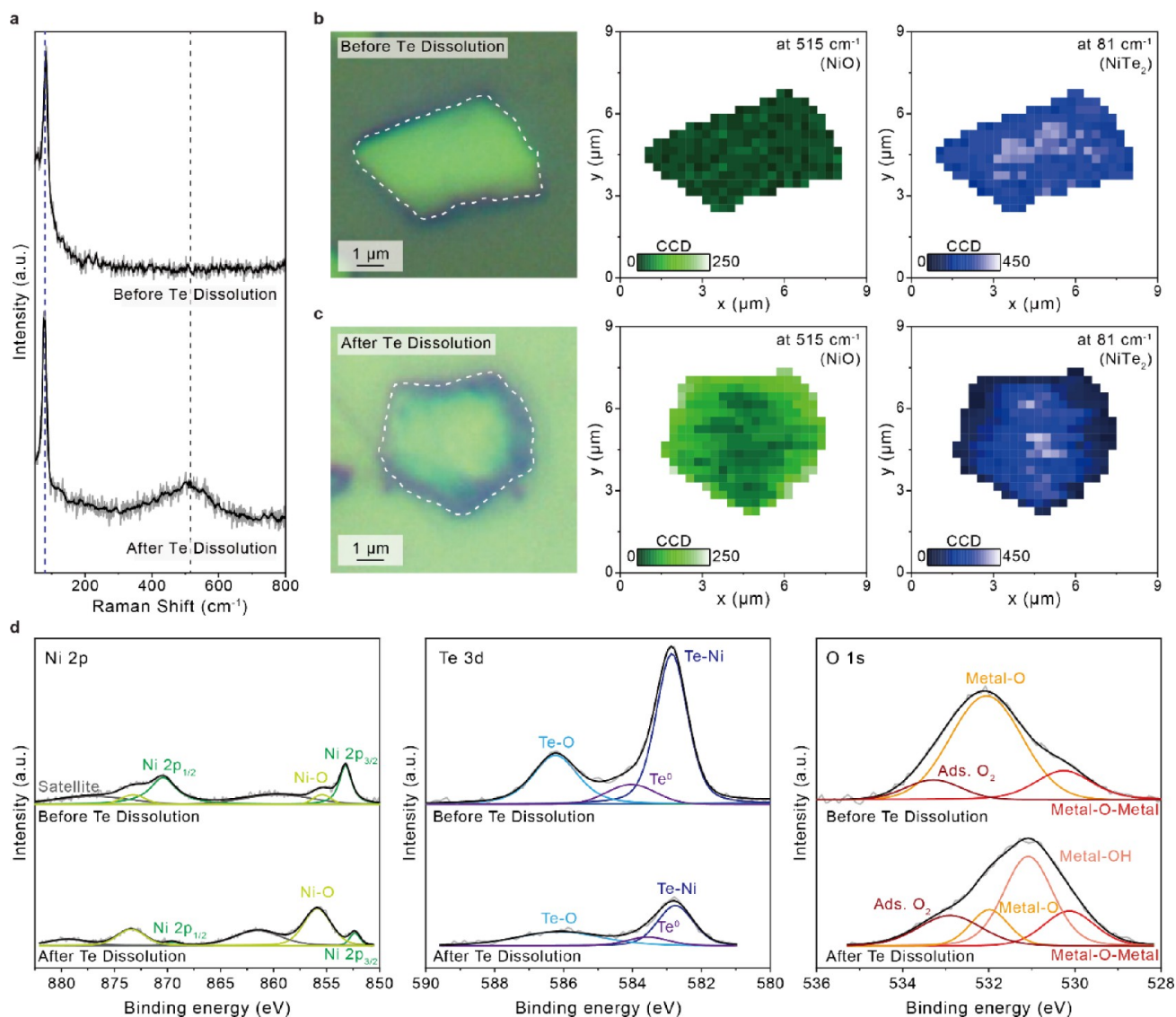


Figure 2. Spectroscopic understanding of $\text{NiTe}_{2-x}\text{O}_x/\text{NiO}_x$ formation by Te dissolution and oxidation. (a) Raman spectra of NiTe_2 before and after Te dissolution (and oxidation). (b–c) Raman maps of the NiTe_2 and resulting $\text{NiTe}_{2-x}\text{O}_x/\text{NiO}_x$ at the single flake level. (d) XPS Ni 2p, Te 3d, and O 1s spectra of the NiTe_2 and the $\text{NiTe}_{2-x}\text{O}_x/\text{NiO}_x$.

steps.³⁰ The (re)activity and stability of these hybrid bifunctional catalysts containing the interfaces of TMDs/metal hydroxides, however, have exhibited inferior performance compared to Pt- and Ir/Ru-based materials.^{30–36}

RESULTS AND DISCUSSION

Here, we demonstrate a fundamentally distinct advancement beyond prior strain-engineering approaches for layered TMDs—which have been largely limited to acidic HER—by achieving true alkaline bifunctionality, enabling both HER and OER through strain-modulated NiTe_2 -derived heterostructures. Specifically, we form $\text{NiTe}_{2-x}\text{O}_x/\text{NiO}_x$ catalysts that operate through dual-site and spillover-assisted pathways, enabling highly efficient alkaline water electrolysis. These bifunctional electrocatalysts were synthesized via electrochemical Te dissolution from mechanically exfoliated NiTe_2 flakes, followed by mild oxidation on prestrained thermoplastic substrates coated with thin gold films. Controlled release of the prestrain generated uniaxial tensile and compressive strains through spontaneously formed microwrinkles, as confirmed by

Raman spectroscopy. X-ray and ultraviolet photoelectron spectroscopy analyses indicate that such strain engineering modulates the surface electronic structures of $\text{NiTe}_{2-x}\text{O}_x/\text{NiO}_x$, enhancing charge transfer and catalytic activity. The optimally strained heterostructures exhibited alkaline HER and OER performances surpassing Pt and IrO_x benchmarks. Machine-learning interatomic potential and density functional theory calculations revealed that Te-vacancy/O-substituted NiTe_2 domains enable oxygen intermediate spillover between adjacent active sites for OER, while strained NiTe_2 domains facilitate hydrogen transfer to NiO with Ni vacancies for HER, collectively reducing free-energy barriers in the multistep electrochemical reactions.

Figure 1 depicts the formation of $\text{NiTe}_{2-x}\text{O}_x/\text{NiO}_x$ heterostructures based on multilayer NiTe_2 crystals. We first transferred mechanically exfoliated NiTe_2 flakes to gold (Au) thin films deposited on prestrained thermoplastic polystyrene (PS) with an average surface coverage of 41.1% in a stepwise manner to achieve sample-to-sample consistency (Figure 1a, Figures S1–S3, and Methods section). Statistical analyses of

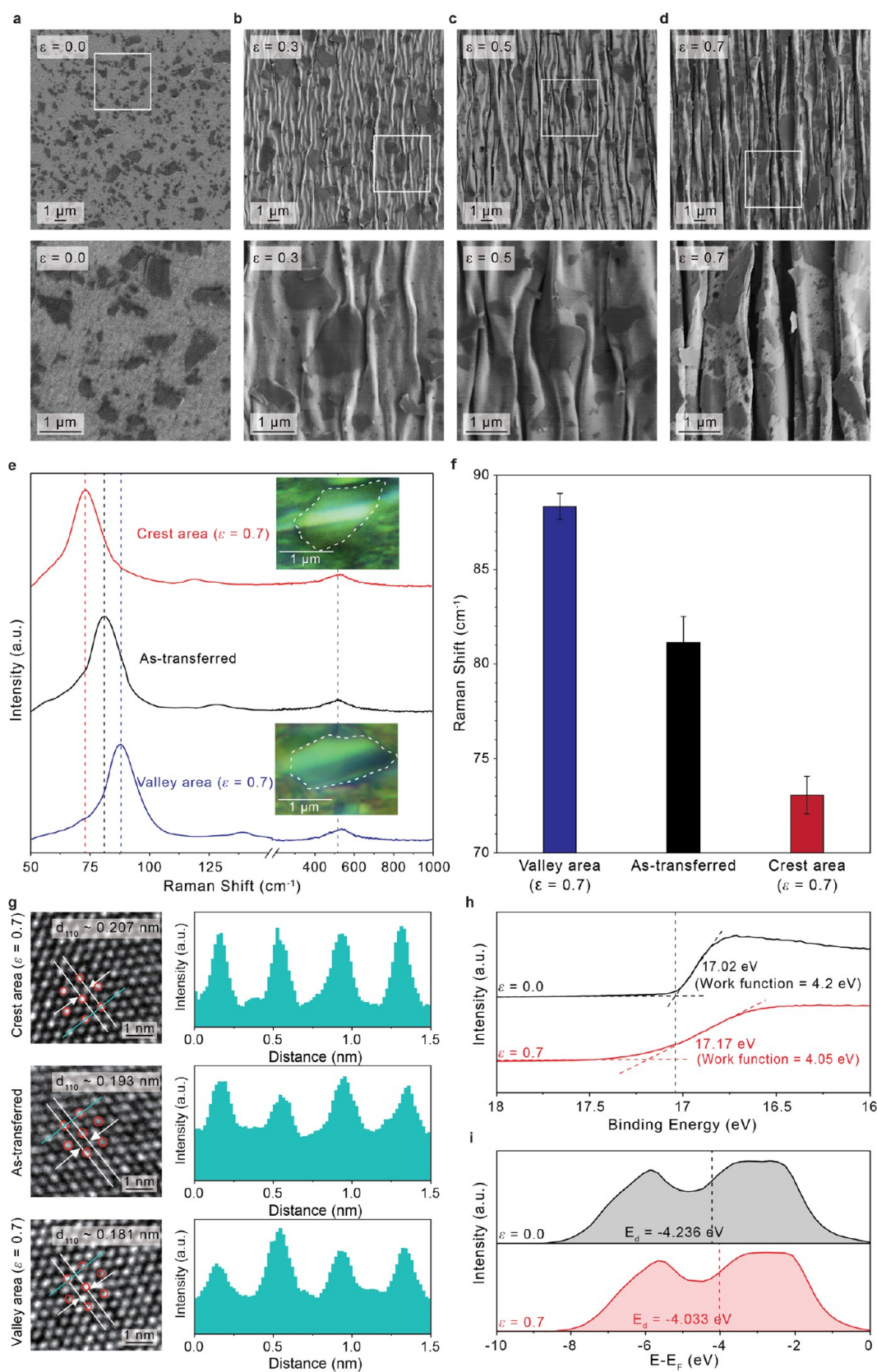


Figure 3. Strain-engineering $\text{NiTe}_{2-x}\text{O}_\delta/\text{NiO}_x$ heterostructures by surface buckling. SEM images of $\text{NiTe}_{2-x}\text{O}_\delta/\text{NiO}_x$ flakes (a) before and after strain engineering at (b) $\epsilon = 0.3$, (c) $\epsilon = 0.5$, and (d) $\epsilon = 0.7$. (e–f) Raman shifts of $\text{NiTe}_{2-x}\text{O}_\delta/\text{NiO}_x$ before and after straining at $\epsilon = 0.7$ over crest and valley areas. (g) Lattice fringes of NiTe_2 domains constituting $\text{NiTe}_{2-x}\text{O}_\delta/\text{NiO}_x$ heterostructures and the corresponding contrast intensity line profiles (marked with cyan lines in the lattice fringes) at crest regions, flat regions without strain, and valley regions under compressive strain. (h) Work function ($\Delta E = E_f - E_{\text{VBM}}$) analyses of $\text{NiTe}_{2-x}\text{O}_\delta/\text{NiO}_x$ before and after strain engineering based on UPS measurements. (i) The d-band center shift induced by strain-engineering measured by valence band XPS. Insets in (e) show optical

Figure 3. continued

images of $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ flakes over the crest and valley regions of buckles. The Raman shifts shown in (f) were measured and averaged over approximately 90 flakes. The penetration depth of the Raman spectroscopy was approximately $2\ \mu\text{m}$, allowing detection of the strains developed throughout the entire volume of the catalysts.

the thickness and size distributions of the as-transferred NiTe_2 , performed by atomic force microscopy (AFM), showed that flakes with an average thickness of $\sim 20\ \text{nm}$ and lateral length of $\sim 0.7\ \mu\text{m}$ were transferred onto the substrates (Figure S4). We then connected copper wires to the Au films to fabricate working electrodes (WEs) for water electrolysis, in a three-electrode configuration consisting of a Pt foil counter electrode (CE) and a Hg/HgO reference electrode (RE) with an alkaline electrolyte of 1 M KOH (Figures S5–S6, Methods section). By applying reductive cyclic voltammetry (CV) over the WE, we formed surface Te vacancies in a controlled manner from NiTe_2 flakes via electrochemical Te dissolution (Figures S7–S8, Methods section).^{37,38} After Te dissolution, we removed the WE from the electrochemical cell and then $\text{NiTe}_{2-\delta}$ flakes were oxidized at $60\ ^\circ\text{C}$ in a convection oven to form $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ heterostructures (Methods section). We checked that the thickness and size distributions of the flakes were remained statistically consistent across arbitrary samples, which indicated that the electrochemical Te dissolution did not induce severe delamination of the materials with sample-to-sample reproducibility (Figure S4, Table S1).

We performed transmission electron microscopy (TEM) to resolve the heterostructures at the atomic scale (Figures 1b–d and Figures S9–S11). The high-resolution TEM (HRTEM) image in Figure S9 shows clear atomic fringes and corresponding fast Fourier transform (FFT) patterns of as-transferred flakes confirming the crystalline 1T- NiTe_2 structure as reported previously.^{39,40} Energy dispersive X-ray (EDX) maps verified a stoichiometric NiTe_2 composition with minute amounts of oxygen, attributed to native oxide on the outer edges as well as chemisorbed oxygen species occupying Te surface vacancies (Figure S10); these surface chalcogen defects have been commonly reported in mechanically exfoliated NiTe_2 .^{40,41} HRTEM images and EDX maps of the flakes acquired after electrochemical Te dissolution and subsequent oxidation reveal drastic structural changes leading to the formation of the heterostructures (Figures 1b–d and Figure S11): (1) the presence of NiTe_2 in the central (core) region of the multilayered flakes; (2) partial conversion of NiTe_2 to $\text{NiTe}_{2-\delta}\text{O}_\delta$ domains distributed near the boundaries of the core NiTe_2 region; and (3) the formation of NiO_x over the outer edges and basal planes of the flakes. Based on the FFT pattern of the central region (yellow box in Figure 1c), we conclude that the 1T- NiTe_2 domains remain predominantly in their original phase throughout the thickness of the flake with the lateral length of $\sim 0.7\ \mu\text{m}$ (i.e., the average size of the transferred flakes). Furthermore, the $\text{NiTe}_{2-\delta}\text{O}_\delta$ domains exhibit atomic fringes that suggest the underlying crystallinity of NiTe_2 is preserved as a structural template. The outer edges (green box in Figure 1c) exhibit the polycrystalline NiO_x matrix with the occasional presence of amorphous domains, indicative of local structural disorder as evidenced by the corresponding FFT patterns. The formation of crystalline NiO is thermodynamically more preferred than its amorphous phase on Ni-based chalcogenides with electrochemical redox processes.⁶ We observed that the ratio of amorphous phase to

polycrystalline areas was much larger for flakes with a lateral length over a few micrometers (Figure S11).

Raman scattering and X-ray photoelectron spectroscopy (XPS) measurements elucidate the formation process of $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ heterostructures (Figure 2). The Raman spectra of as-transferred NiTe_2 flakes exhibited a characteristic vibrational mode at $\sim 81\ \text{cm}^{-1}$ corresponding to the 1T phase of NiTe_2 (Figure 2a). Following Te dissolution and subsequent oxidation, a new vibrational mode emerged at $\sim 515\ \text{cm}^{-1}$ associated with NiO_x , indicating the coexistence of $\text{NiTe}_{2-\delta}\text{O}_\delta$ and NiO_x domains.^{42,43} To visualize the spatial distributions of the $\text{NiTe}_{2-\delta}\text{O}_\delta$ domains (maintaining 1T- NiTe_2 crystallinity) and the mixed amorphous and crystalline NiO_x in the outer regions, we performed Raman mapping analyses (Figures 2b–c). These maps were constructed using the intensity of the NiO_x longitudinal optical phonon mode ($\sim 515\ \text{cm}^{-1}$) and the E_g phonon mode of 1T- NiTe_2 ($\sim 81\ \text{cm}^{-1}$). The as-transferred NiTe_2 flakes exhibited a strong and uniform 1T- NiTe_2 signal across their entire surfaces, with negligible intensity of the NiO_x mode, indicating minimal surface oxidation prior to treatment. We also note that the signal intensity of the E_g phonon mode is slightly higher in the central regions of the flake, likely due to reduced native oxide coverage in comparison to the outer edges. After Te dissolution and oxidation, we observed a significant increase in the intensity of NiO_x , particularly over the flake edges, where the intensity of 1T- NiTe_2 signal was reduced. In contrast, the central regions retained a strong 1T- NiTe_2 signal, indicating that the core domains remained largely intact. These spatially resolved Raman data suggest that the transformation from 1T- NiTe_2 to $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ initiates at the edges and progresses inward toward the central regions. The preservation of 1T- NiTe_2 features in the core, coupled with the emergence of NiO_x signals at the periphery, confirms the formation of $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ heterostructures through controlled electrochemical reconstruction.

We performed compositional analyses of as-transferred NiTe_2 flakes and the resulting $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ heterostructures using XPS (Figure 2d). The pristine flakes prior to Te dissolution exhibited the core level Ni 2p (at 853.2 and 870.4 eV) and the Te 3d (at 582.9 eV) peaks characteristic of the 1T- NiTe_2 phase, matching with the Raman scattering data.^{40,44} The peak at 586.3 eV of Te 3d spectrum indicates the presence of native oxides formed via Te–O bonding, as further supported by the O 1s spectrum showing features at 532.1 and 530.3 eV attributed to Metal–O and Metal–O–Metal bonds, respectively.^{40,44} Following the electrochemical treatment, Ni 2p spectra shows a marked reduction in the NiTe_2 related peaks (at 852.3 and 869.6 eV) and a corresponding increase in the NiO_x peak (at 855.9 and 873.4 eV), verifying the formation of $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ heterostructures. Simultaneously, the Te 3d peaks associated with Te–O and Te–Ni bonding reduced significantly, consistent with the dissolution of Te atoms. Quantitative analysis revealed that approximately $\sim 87\%$ of Te atoms were removed from as-transferred flakes (Figure S12). The O 1s spectrum following the formation of heterostructures exhibited an additional feature at 531.1 eV corresponding to

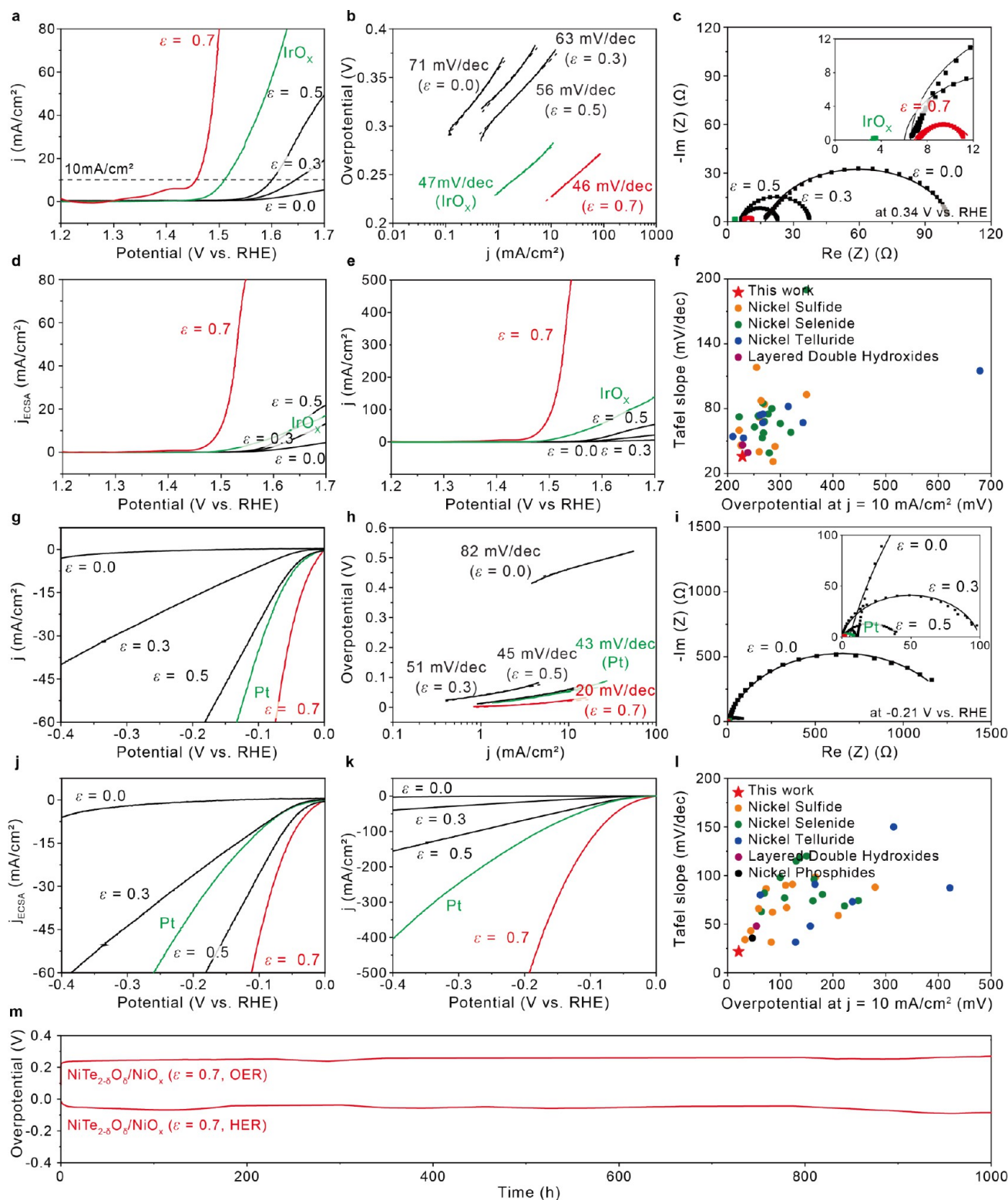


Figure 4. OER/HER performance of strain-engineered $\text{NiTe}_{2-x}\text{O}_\delta/\text{NiO}_x$ heterostructures. (a) iR -corrected OER polarization curves and (b) corresponding Tafel plots from the electrochemical cells containing $\text{NiTe}_{2-x}\text{O}_\delta/\text{NiO}_x$ before ($\epsilon = 0.0$) and after strain-engineering ($\epsilon = 0.3$, $\epsilon = 0.5$, and $\epsilon = 0.7$) in comparison with IrO_x . (c) Nyquist plots of the electrochemical cells and (d) the polarization curves normalized to the ECSA, i.e., normalized current density to ECSA (j_{ECSA}) versus potential. (e) The OER polarization curves over high current density regimes. (f) OER performance of strained $\text{NiTe}_{2-x}\text{O}_\delta/\text{NiO}_x$ at $\epsilon = 0.7$ in comparison to reported (Ni-based) transition metal chalcogenides detailed in Table S1. (g) iR -corrected HER polarization curves and (h) corresponding Tafel plots from the electrochemical cells containing the heterostructures in comparison with Pt. (i) Nyquist plots of the corresponding electrochemical cells, (j) polarization curves normalized to the ECSA, and (k) HER polarization curves over high current density regimes. (l) HER performance of the strain-optimized heterostructure in comparison to reported (Ni-based) transition metal chalcogenides summarized in Table S2. (m) The stability of alkaline OER/HER from the heterostructure using chronopotentiometry at 10 mA/cm^2 .

nickel hydroxides (Metal–OH) alongside the oxygen-related species associated with local $\text{NiTe}_{2-\delta}\text{O}_\delta$ and NiO_x phases.

The electronic structures of layered TMDs, in particular, Dirac semimetals such as NiTe_2 and PtSe_2 , are tunable with strain-engineering, and consequently the design of the electrocatalysts with enhanced charge transfer has been shown to be possible.^{23,45} We uniaxially relieved the prestrain by heating the PS substrates above the PS glass transition temperature (T_g) to generate ordered Au microwrinkles with buckled $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ flakes, where the amount of macroscopic strain (ϵ) is tunable by varying the heating time as well as the level of prestrain relaxation (Figures 3a–d, Figure S5, Methods section). Here, we mechanically clamped each end of the substrates before heating to induce the uniaxial strains. Wrinkles formed across the entire surface of the centimeter-scale samples in parallel because of the difference between Young's modulus of the skin layer (in this case, the Au film partially covered with $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$) and the substrates.^{46–50} The scanning electron microscopy (SEM) images showed that the flat surfaces of Au films with flakes transformed to three-dimensional (3D) microwrinkles after straining such that flakes with a lateral size larger than a few hundred nanometers buckled out-of-plane. As ϵ increased, the buckled flakes were confined more tightly between adjacent Au wrinkles because the wrinkling wavelength decreased with strain.^{21,22}

To characterize the lattice strains developed in the buckled $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ heterostructures, we analyzed Raman shifts at the single flake level (Figures 3e–f). Here, we selected the flakes with lateral size of a few micrometers to fully collect spatially resolved information considering the spot size of the Raman laser (Methods section). The formation of buckles upon applied macroscopic strain ($\epsilon = 0.7$) spontaneously gives rise to “valley” regions with compressive strains and “crest” regions with tensile strain, as previously reported.⁵¹ The E_g phonon mode of NiTe_2 exhibited shifts in response to these strained states, where the crest (valley) region indicates a shift to lower (higher) wavenumbers by $\sim 74\text{ cm}^{-1}$ ($\sim 89\text{ cm}^{-1}$), suggesting internal tensile (compressive) strain values of 5.09% (5.02%) within the flakes on average induced by the 3D wrinkling.^{51–53} We further note that the magnitude of both tensile and compressive strain is tunable by adjusting the applied prestrain ϵ , as verified by the corresponding Raman peak shifts (Figure S13). In contrast, the phonon mode of NiO showed negligible spectral changes, with no discernible shifts under applied strains. This insensitivity to strain is likely attributed to the structural nature of the material, consisting of randomly oriented nanoscale polycrystalline domains interspersed with amorphous regions. Such structural disorder limits the Raman signal quality and complicates the reliable detection of phonon mode variations typically associated with lattice strains. The contrast-intensity line profiles from the HRTEM images further confirmed that the $\text{NiTe}_{2-\delta}\text{O}_\delta$ domains with NiTe_2 (110) plane, which has an intrinsic lattice constant of 0.386 nm,⁵⁴ exhibited a locally expanded lattice constant of $\sim 0.415\text{ nm}$ under tensile strain at the crest regions (corresponding to $\sim 7\%$ tensile strain) (Figure 3g and Figure S14). In contrast, the compressed valley regions showed a reduced lattice constant of $\sim 0.362\text{ nm}$ (corresponding to $\sim 6\%$ compressive strain). Ultraviolet photoelectron spectroscopy (UPS) analyses of the secondary cutoff revealed a decrease in the work function of the heterostructures by $\sim 0.15\text{ eV}$ (on average) with $\epsilon = 0.7$ (Figure 3h and Methods section). The

valence band spectra obtained by XPS (VB-XPS) further confirms the enhanced metallic behavior of the strained heterostructures; the density of states moves closer to the Fermi level (E_f) with strain and the d band center shifted upward by $\sim 0.2\text{ eV}$ (on average) from the same samples used for the UPS measurements (Figure 3i and Methods section). These strain-induced modulations of the electronic structures are expected to enhance the surface electrochemical reactions as we demonstrate in the following sections regarding OER/HER measurements and theoretical free energy values for the adsorption of proton and oxygen species. We note that statistical analysis revealed an approximately 1:1 population ratio of heterostructures located at crest and valley regions, indicating that both compressive and tensile strains contributed comparably to modulating the electronic structures and the resulting OER/HER activities (Figure S15).

We experimentally quantified how the formation of heterostructures and their strain engineering enhance the alkaline OER/HER activities (Figure 4, Methods section). The measured polarization curves show that the activity of as-transferred NiTe_2 was almost negligible with an onset overpotential (at low current density, j , of 0.1 mA/cm^2) of 280 mV (i.e., the cell potential of 1.51 V vs the reversible hydrogen electrode (RHE), for which the OER activity is similar to bare Au films (Figure S16). After Te dissolution and oxidation, both the onset overpotential and the overpotential at j of 10 mA/cm^2 decreased to 238 mV and 580 mV, respectively, indicating improved electrocatalytic activity. As we apply ϵ increasing to 0.7, the overpotentials decreased dramatically to 60 mV and 228 mV with electrocatalytic activity greater than IrO_x (Figure 4a). Figure 4b and Figure S16 highlight that the corresponding Tafel slopes decreased from 94 mV/dec for pristine NiTe_2 to 71 mV/dec ($\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$) to 46 mV/dec ($\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ with ϵ of 0.7), which is favorable compared to the measured value of 47 mV/dec for IrO_x . We performed electrochemical impedance spectroscopy (EIS) analyses to check that the OER activity of $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ was higher than that of NiTe_2 due to the suppressed charge transfer resistance (R_{CT}) (Figure 4c and Figures S16–S17). The R_{CT} of the cell was further reduced by more than 20-fold after straining, which implies the enhanced electron transfer between heterostructures and OH^- ions in the electrolyte solution to form the initial intermediate species. Figure S18 shows that straining the as-transferred NiTe_2 enhances its catalytic activity. Although the OER performance of bare NiTe_2 increases with strain up to $\epsilon = 0.7$ —mainly due to reduced charge-transfer resistance—it still remains far lower than that of the strained $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ heterostructures. We note that the OER performance reached its maximum at 30 CV cycles, corresponding to the formation of Te vacancies. Additional cycling led to catalyst delamination, resulting in a subsequent decrease in catalytic activity (Figures S19–S20).

The electrochemically active surface area (ECSA) for each cell was estimated using the electrochemical double-layer capacitance (C_{DL}), the value of which was determined by measuring the non-Faradaic capacitive current associated with double-layer charging from the scan-rate dependence of cyclic voltammetry (CV); the C_{DL} of $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ was enhanced by 350% with straining (Figure S21). Assuming that the C_{DL} is proportional to ECSA, this result confirms that strain-engineering of the heterostructures maximized the density of catalytically active sites for oxygen evolution.^{55,56} Figure 4d

shows the polarization curves normalized to the corresponding C_{DL} and the relative configurations of the curves (i.e., the current density per active area, j_{ECSA} , as a function of overpotential) matched with the overall trends in Figure 4a. The larger gap in OER activity per active area between $NiTe_{2-\delta}O_\delta/NiO_x$ and IrO_x resulted because we prepared the WE of IrO_x based on carbon cloth (CC), which has a larger surface area than that of the Au films (Figure S22). Figure 4e shows that the OER performance gap tends to increase under the j regime up to 500 mA/cm²; for example, at an overpotential of 310 mV, the electrochemical cell containing IrO_x exhibited only 4% of the j measured for the WE with the heterostructures. Figure 4f and Figure S23 compare this OER performance of $NiTe_{2-\delta}O_\delta/NiO_x$ with electrocatalysts containing Ni based TMDs, in terms of Tafel slopes and required overpotentials, summarized in Table S2.

On the top of the OER greater than IrO_x , the strain-engineered catalysts outperform the pure Pt in alkaline HER. Figure 4g shows the HER polarization curves of $NiTe_{2-\delta}O_\delta/NiO_x$ before and after straining at varying ϵ in comparison with Pt foil. We observed enhanced HER performance of $NiTe_{2-\delta}O_\delta/NiO_x$ as ϵ increased to 0.7; the overpotential at the j of 10 mA/cm² was suppressed from −0.462 (before straining) to −0.024 (V vs RHE) with ϵ . Considering that the potential required to achieve the same j using Pt was measured as −0.054 (V vs RHE), ~2 times higher than that for the strained heterostructures, our approach enables efficient alkaline HER operating over ultralow overpotential regimes. Although the HER performance of bare $NiTe_2$ shows a modest strain-induced improvement due to reduced charge-transfer resistance, its activity remains substantially lower than that of the strained $NiTe_{2-\delta}O_\delta/NiO_x$ heterostructures, consistent with the OER trend (Figure S18). We characterized the HER kinetics of $NiTe_{2-\delta}O_\delta/NiO_x$ modified with ϵ by plotting Tafel slopes based on the corresponding polarization curves (Figure 4h). The Tafel slope before straining was measured with ~82 mV/dec, indicating that both the Volmer (Tafel slope of ~120 mV/dec) and Heyrovsky (Tafel slope of ~40 mV/dec) reactions could be the rate-determining step.⁵⁷ The reaction kinetics of the catalysts changed after straining at ϵ of 0.7, with a significantly reduced Tafel slope of ~20 mV/dec (lower than ~43 mV/dec of Pt), in which the Tafel reaction mechanism should govern HER and the desorption of hydrogen was the rate-limiting step after water dissociation. We note that the HER activity of $NiTe_2$ before Te dissolution was negligible (Figure S16). Similar to the OER results, the HER activity was also maximized at 30 CV cycles, which enabled the formation of Te vacancies without causing catalyst delamination (Figures S20 and S24).

The R_{CT} of the electrochemical cell containing $NiTe_{2-\delta}O_\delta/NiO_x$ catalysts over the reductive potentials for alkaline HER was decreased by approximately 3 orders of magnitude with ϵ , revealing that the electron transfer between the heterostructures and the protons in the electrolyte solution increased and resulted in reduced Tafel slopes (Figure 4i and Figure S25). We observed that the decreases in R_{CT} were partially attributable to substrate structuring effects with improved surface wettability and mass transport both for OER and HER (Figure S26). Figure 4j shows the HER polarization curves normalized by ECSA to compare the per-site activities of strained heterostructures with Pt; the resulting activity of the $NiTe_{2-\delta}O_\delta/NiO_x$ with $\epsilon > 0.5$ started to outperform Pt. This efficient alkaline HER further induced a higher j up to 500

mA/cm² at −0.192 (V vs RHE), where only 23% of the current density of the heterostructured electrode was measured from the cell with a Pt WE under the same conditions (Figure 4k). We also show that the resulting HER from strained $NiTe_{2-\delta}O_\delta/NiO_x$ exceeds the performance of most nickel-based electrocatalysts in alkaline electrolytes (Figure 4l, Figure S27, Table S3). We note that applying strain levels up to $\epsilon \approx 0.8$ led to decreased HER activity due to cracking of the underlying Au films, a trend that was similarly observed for the OER (Figure S28).

We tested the durability of the bifunctional electrocatalysts with chronopotentiometry measurements and demonstrated stable OER/HER for 1000 h without notable performance degradation (Figure 4m, Figure S29). The internal compressive and tensile strains were retained after the stability tests, as statistically confirmed by Raman spectroscopy (Figure S30). The same measurements were performed for the comparison groups with IrO_x and Pt WEs, and we observed less stability from these systems (Figure S31). We further measured the gaseous byproducts from the electrochemical cells with strained heterostructures, and near-unity Faradaic efficiencies (~99.8 and ~98.7% for OER and HER, respectively) were recorded (Figure S32). To this end, we constructed a laboratory-scale two-electrode electrolyzer cell based on $NiTe_{2-\delta}O_\delta/NiO_x$ heterostructures to verify enhanced alkaline water electrolysis (Figure S33). This full cell consists of two electrodes with $NiTe_{2-\delta}O_\delta/NiO_x$ electrocatalysts as both an anode and a cathode (denoted as $NiTe_{2-\delta}O_\delta/NiO_x$ - $NiTe_{2-\delta}O_\delta/NiO_x$ cell). This cell exhibited a cell potential of 1.67 V at a current density of 100 mA/cm², which is lower than the potential of 1.74 V for a Pt- IrO_x cell (i.e., a comparison system composed of Pt foil as a cathode and IrO_x/CC as an anode), indicating that the bifunctionality from the heterostructures are indeed critical for improved overall water splitting. We observed performance gaps between these two cells, particularly in the high current density regime with the $NiTe_{2-\delta}O_\delta/NiO_x$ - $NiTe_{2-\delta}O_\delta/NiO_x$ cell demonstrating a current density of up to ~1500 mA/cm² at a potential of 2.01 V, while only 25% of the current density level (i.e., 377 mA/cm²) was generated for the Pt- IrO_x cell at the same potential. Current density levels exceeding 1 A/cm² suggest that the system can be utilized for the mass production of molecular hydrogen for industrial-scale applications.^{58–60} Furthermore, we verified that three-electrode cells employing $NiTe_{2-\delta}O_\delta/NiO_x$ catalysts sustain such high current densities for both the OER and HER, thereby enabling efficient overall water electrolysis comparable to commercially available systems (Figure S34, Table S4). We characterized the durability of the electrolyzer containing the heterostructures by chronopotentiometry, highlighting the long-term stability at low and high current levels without severe performance degradation for 750 h of operation. Since the heterostructures exhibit mass activities for both the OER and HER that are comparable to those of state-of-the-art noble-metal-based electrocatalysts (Figure S35, Tables S5–S6), this electrolyzer system has strong potential to be further developed into a membrane-electrode-assembly type electrolyzer. Furthermore, the overall bifunctional performance of $NiTe_{2-\delta}O_\delta/NiO_x$ remained consistent even in much higher electrolyte concentrations, a characteristic that is critical for industrial adoption (Figure S36).

We explored the OER/HER enhancement mechanisms from $NiTe_{2-\delta}O_\delta/NiO_x$ heterostructures using machine-learning

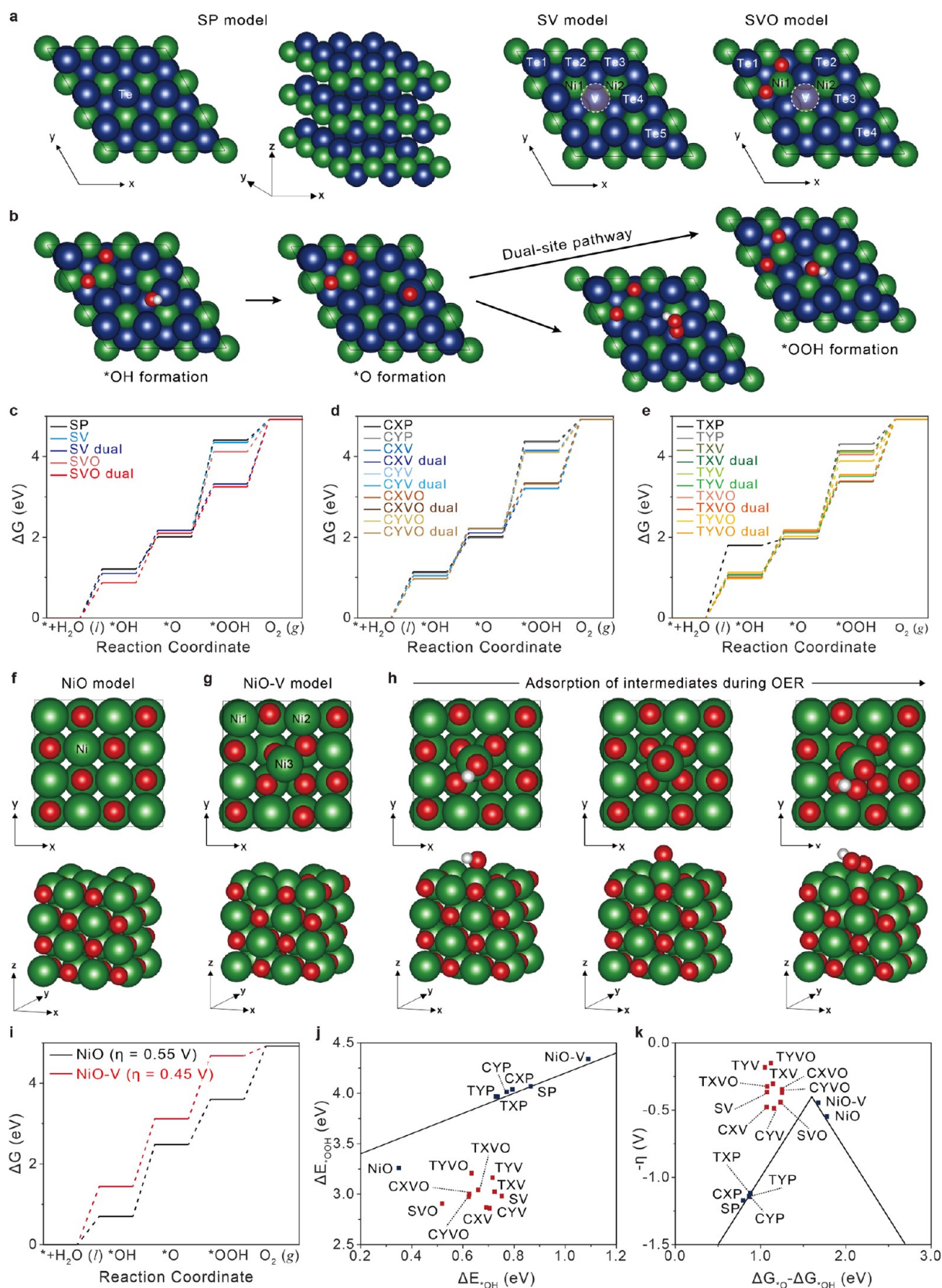


Figure 5. Theoretical understanding of alkaline OER from $\text{NiTe}_{2.6}\text{O}_8/\text{NiO}_x$ heterostructures. (a) Modeled surfaces of SP, SV, and SVO domains for DFT calculations. (b) Schematics representing single- and dual-site OER pathways on the SVO model. Both the formation of initial $^*\text{OH}$ and subsequent $^*\text{O}$ intermediates were confined to the primary atomic site (Te3) and the spillover of the intermediate along the dual-site pathway resulted in $^*\text{OOH}$ formation on an adjacent Ni2 site. (c–e) Calculated Gibbs free energy diagrams for all the domain models with single and dual-site mechanisms. Modeled surfaces of (f) NiO and (g) NiO-V domains, and (h) surface reconstruction processes of NiO-V during OER. In (h), all the intermediates formed on the primary Ni3 site with a single-site mechanism. (i) Calculated Gibbs free

Figure 5. continued

energy diagrams for OER from NiO and NiO-V models. (j) Scaling relation between ΔE_{OOH}^* vs ΔE_{OH}^* and (k) volcano plot correlating OER overpotential versus $\Delta G_{\text{O}}^* - \Delta G_{\text{OH}}^*$.

interatomic potential (MLIP) and density functional theory (DFT) calculations (Figure 5). Here, we constructed the supercell models depicting NiTe_2 , $\text{NiTe}_{2-\delta}$, $\text{NiTe}_{2-\delta}\text{O}_\delta$, and NiO_x domains, which compose the surfaces of heterostructures. The models based on crystalline 1T- NiTe_2 were labeled with the conditions of directional strains (S: standard condition, i.e., before straining; C: compressive strains; T: tensile strains; X: strains applied parallel to x axis; Y: strains applied parallel to y axis) and the stoichiometry of surfaces (P: pristine surfaces; V: surfaces with Te vacancies; VO: surfaces with Te vacancies and O substitutions) (Figure 5a, Figures S37–S39, Methods section). We considered the conventional four-step processes in the OER through the formation of $^*\text{OH}$, $^*\text{O}$, and $^*\text{OOH}$ intermediates on model surfaces with associated free energy barriers ($\Delta G_1 \sim \Delta G_4$, Methods section). All the adsorption sites were evaluated using MLIP calculations, followed by DFT analyses for the most reactive site from each model (Figure S40). On the domain models with pristine surfaces before and after straining (i.e., SP, CXP, CYP, TXP, TYP), a single available adsorption site confined the entire reaction processes to a single-site mechanism (Table S7). In contrast, the model surfaces with Te vacancies and O substitutions (i.e., SV, SVO, CXV, CYV, CXVO, CYVO, TXV, TYV, TXVO, TYVO) provide dual-site pathways in which two adjacent, nonequivalent atomic sites enable the spillover of oxygen intermediates during the OER (Figure 5b, Tables S8–S9). In such pathways, we computationally observed that $^*\text{OH}$ intermediates were adsorbed and remained at their primary sites until the formation of $^*\text{O}$ intermediates, and then migrated along the most favorable pathway to form $^*\text{OOH}$. For each model, we set the dual-site pathways within the spillover constraint (<3.5 Å) by considering the free energy barriers with corresponding overpotentials to form $^*\text{OOH}$ (Tables S4–S5, Methods section).^{61–66} During OER, the intensity of the Raman band corresponding to $^*\text{OOH}$ species adsorbed on Ni active sites (Ni–OOH band) increased progressively with increasing overpotential,^{67,68} accompanied by a simultaneous enhancement of the Te–O band intensity (Figure S41). This observation suggests that the initially formed $^*\text{OH}$ intermediates adsorbed on Te sites within the $\text{NiTe}_{2-\delta}\text{O}_\delta$ domains are subsequently converted to $^*\text{O}$ species and migrate to neighboring Ni sites to form $^*\text{OOH}$, supporting the theoretical dual-site mechanism.

Figures 5c–e show the resulting ΔG diagrams in alkaline OER calculated from the supercell models. The rate-determining step (RDS) for OER on the models with a single-site mechanism was the third step to form $^*\text{OOH}$ with ΔG_3 , whereas the reactions with the dual-site pathways stabilized the intermediate formation sufficiently such that the theoretically required OER overpotentials (η) were reduced with balanced four step processes (Tables S8–S9, Figure S42). The surface of TYVO exhibited the most significant decrease in η (from 0.64 to 0.14 V) by incorporating the dual-site mechanism, followed by TXVO (from 0.67 to 0.32 V), CXVO (from 0.64 to 0.35 V), and CYVO (from 0.64 to 0.37 V). Considering the theoretical η values of 0.4–0.6 eV for IrO_2 from the literature,^{69–71} these

strains applied to the constituent $\text{NiTe}_{2-\delta}\text{O}_\delta$ (and $\text{NiTe}_{2-\delta}$) domains were the dominant factor to facilitate the intermediate adsorptions to achieve OER performance greater than IrO_x . We analyzed the charge density differences and Bader charges to reveal the stabilization process of $^*\text{OOH}$; the dual-site mechanism bridged the $^*\text{OOH}$ with exposed Ni sites underneath Te vacancies through doubled electron donations, which gives rise to the larger net charge transfer (Δq) than the cases involving single-site mechanisms (Figures S43–44). In general, the larger Δq with enhanced (localized) electron density causes the lower η of the active sites for OER.⁷² The extended charge analyses of the surface-bound oxygen atoms in $^*\text{OOH}$, for all the model surfaces before and after strain engineering, confirmed the higher Δq values from the reactions with dual-site pathways than the single-site cases (Table S11).

We simulated the OER processes on NiO_x domains by constructing the crystalline NiO and NiO-V with a Ni vacancy (Figures 5f–g, Methods section). We regarded these structures as a system representation rather than an amorphous phase because the average flake size resulted the polycrystallinity-dominant NiO_x , enclosing $\text{NiTe}_{2-\delta}\text{O}_\delta$ domains after Te dissolution and oxidation (Figure 1c). For these models, we did not consider the effects of strains as verified by aforementioned Raman shifts in Figure 3e. Although we observed the surface reconstructions of NiO-V during OER (Figure 5h), which could modify the local coordination environments, the single-site localization of successive intermediate adsorptions ($^*\text{OH}$, $^*\text{O}$, and $^*\text{OOH}$) identified a lower η of 0.45 V compared to NiO (0.55 V) (Figure 5i). Considering the much larger level of $\eta > 1.0$ V from the domains having NiTe_2 stoichiometry (SP, CXP, CYP, TXP, TYP models, Table S7), the formation of heterostructures containing NiO_x domains benefits OER (re)activity regardless of strain engineering. Notably, the RDS with ΔG_2 from NiO was shifted to the third step for the $^*\text{OOH}$ adsorption after forming the vacancy, and the fourth step to form gaseous O_2 could also limit the OER kinetics on the NiO-V as evaluated by the lattice oxygen mechanism (Figure S45). We note that the NiO and NiO-V domains were not considered dual-site pathways for the intermediate spillovers due to limited site separation and strong intermediate bindings on the primary sites (Methods section, Table S10). We assessed overall trends of OER reactivities for all the model surfaces by analyzing the scaling relations between ΔE_{OH}^* and ΔE_{OOH}^* , and found that the surfaces of NiO/NiO-V and pristine NiTe_2 (before and after straining, SP, TXP, TYP, CXP, CYP) followed the conventional linear relation (Figure 5j). Interestingly, the NiTe_2 models with incorporated Te vacancies and O substitutions exhibited significant deviations from the governing relation. The volcano plot correlating the collected η values with corresponding $\Delta G_{\text{O}}^* - \Delta G_{\text{OH}}^*$ (Figure 5k) verifies that these deviations lead to the breakdown of the typical scaling relation, and the design of catalytic activity beyond the theoretical minimum ($\eta \sim 0.37$ V) was thus possible.⁷³

We also evaluated HER (re)activity on the supercell models by calculating water dissociation activation energy (E_a) and the Gibbs free energy of hydrogen adsorption (ΔG_{H}^*), both of

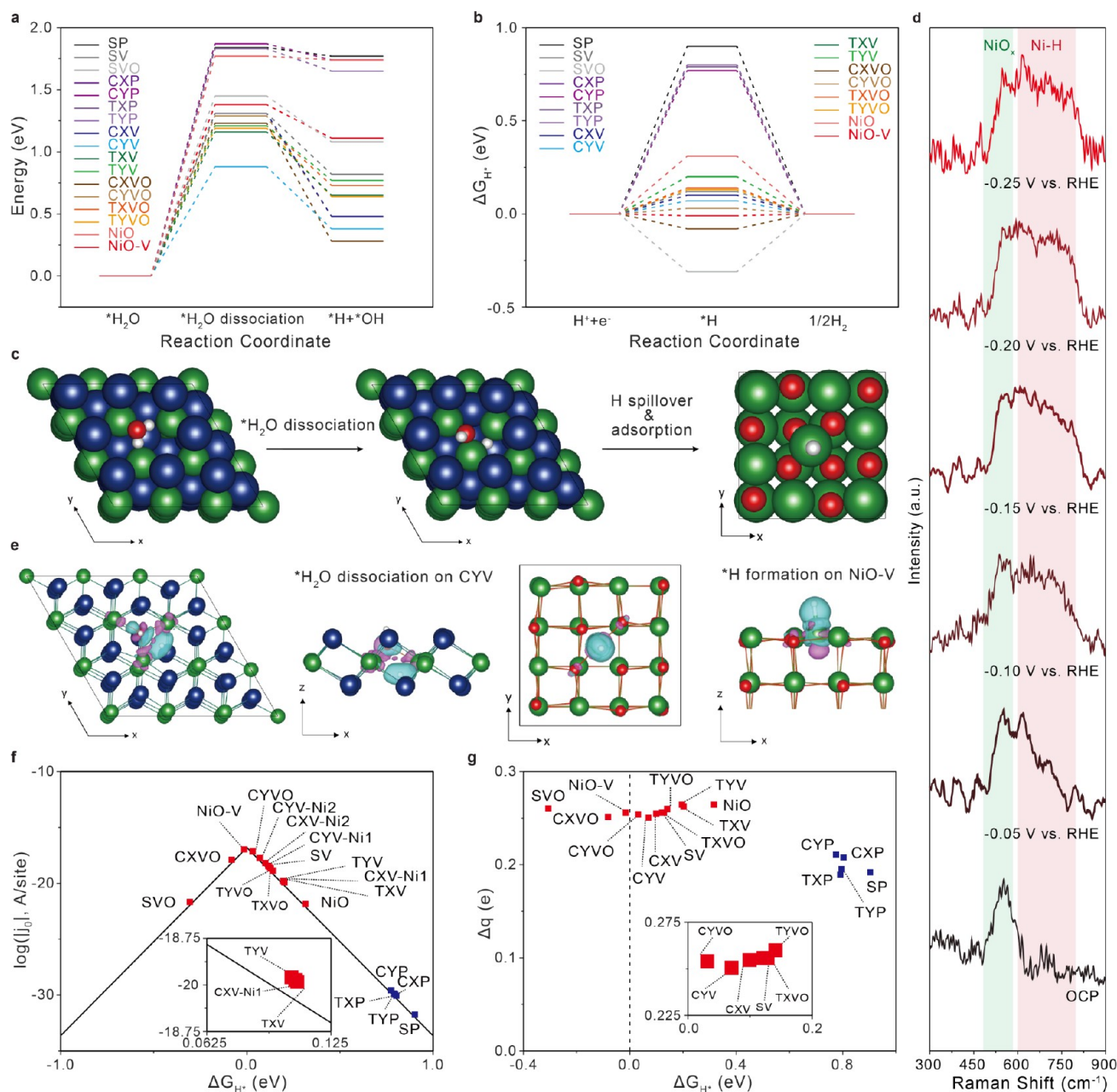


Figure 6. Theoretical understanding of alkaline HER from $\text{NiTe}_{2.8}\text{O}_8/\text{NiO}_x$ heterostructures. Calculated (a) E_a and (b) ΔG_{H} values for all the domain models in Figure 5. (c) Suggested alkaline HER mechanism starting from water dissociation on CYV domain (Ni1 site) followed by hydrogen spillover to form H intermediate on NiO-V domain (Ni3 site). (d) In situ Raman spectra of the strained heterostructures (at $\epsilon = 0.7$) during HER over varying overpotentials. (e) Charge density differences on the surface of CYV (after water dissociation) and the surface of NiO-V with H formation. Regions with electron accumulation/deficiency were marked with cyan/magenta color. (f) Volcano plot comparing the HER (re)activities of all the active sites from domain models. (g) The relation between Δq and ΔG_{H} showing the amount of electron transfer to form H for all the active sites.

which are the critical descriptors in alkaline conditions.⁷⁴ Since most of the 1T- NiTe_2 based domains with compression and tension exhibited E_a values lower than the pristine surface (SP model), it is reasonable to conclude that the strain engineering accelerated the initiation of the Volmer step ($\text{H}_2\text{O} + * \rightarrow \text{H} + \text{OH}^-$) in alkaline HER (Figure 6a). In particular, the exposed Ni site under the Te vacancy from CYV induced the lowest E_a of 0.88 eV, the value of which is comparable to Pt (0.8 ~ 0.9 eV);⁷⁵ the same surface before straining (SV) resulted in relatively sluggish kinetics with E_a of 1.31 eV. We note that the efficiency of water dissociation on NiO (E_a of

1.77 eV) was also enhanced by forming Ni vacancies (NiO-V, E_a of 1.38 eV). Figure 6b highlights that the incorporation of Te vacancies and O substitutions can bring the hydrogen adsorption ΔG_{H} closer to 0, and the reactivities of SV (ΔG_{H} of 0.12 eV) and SVO (ΔG_{H} of -0.31 eV) further increased with tensile strains, i.e., ΔG_{H} of 0.10 eV (0.07 eV) for CXV (CYV) and 0.08 eV (0.03 eV) for CXVO (CYVO), respectively. Significantly, NiO-V exhibited nearly thermoneutral hydrogen adsorption with ΔG_{H} of -0.01 eV, from which we propose the best scenario for alkaline HER on $\text{NiTe}_{2.8}\text{O}_8/\text{NiO}_x$; the H species formed by CYV domains should migrate

to nearby NiO-V surfaces with more favorable ΔG_{H} via hydrogen spillover, resulting in HER greater than pure Pt (Figure 6c). The hydrogen spillover among metal-deficient and oxygen-rich domains has been shown to be feasible across interdomain distances.^{73–78} During the HER process, the relative Raman intensity of the Ni–H stretching mode ($\sim 600\text{--}800\text{ cm}^{-1}$) increased under more negative overpotentials, indicating enhanced Volmer intermediate adsorption at Ni sites on the NiO_x regions,^{79,80} partially due to hydrogen spillover (Figure 6d). This spillover-assisted mechanism is further supported by HER measurements in H₂O and D₂O electrolytes (Figure S46), where the activity in H₂O was higher and the calculated kinetic isotope effect exceeded 1.⁸¹

Charge density difference analyses provide further insights into the mechanism on how the domain surfaces resulted in superior water dissociation and hydrogen evolution (Figure 6e). We captured the change in the local charge distributions right after splitting the water molecule on CYV, where the exposed Ni atom underneath a Te vacancy pulls an H atom to break the constituent O–H bonding of the molecule such that the resulting *H intermediate is formed on the atomic site. The remaining OH[−] is also adsorbed onto an adjacent Ni atom near the vacancy, mutually stabilizing the adsorbed species. The surface of NiO-V also shows the surface-mediated charge localization for hydrogen adsorption. We visualized the localized charge transfer from the Ni atom (adjacent to vacancy) to the intermediate leaving depletion regions around neighboring surface atoms. The volcano plot correlating atomic scale HER reactivity (ΔG_{*H}) and activity (j_0 , exchange current per site) summarizes the overall HER performance of all the characterized atomic sites (Figure 6f, Methods section). We reiterate that both compressive and tensile strains tend to enhance the atomic scale reactivity as well as activity of NiTe₂-based domains; CYV, CXVO, and CYVO were positioned near the peak of the volcano, approaching the thermoneutral optimum. We also correlated ΔG_{*H} with corresponding Δq values of the active sites to quantify the amount of charge transfer as a function of adsorption energy in HER (Figure 6g). The active sites with $|\Delta G_{*H}| < 0.4\text{ eV}$ exhibited greater electron transfer to H ($\sim 0.26\text{ e}$), whereas the sites with $|\Delta G_{*H}| > 0.7\text{ eV}$ displayed reduced charge transfer ($\sim 0.20\text{ e}$). These results suggest that approximately $\sim 0.26\text{ e}$ of charge transfer to *H represents a quantitative threshold for thermoneutral adsorption, in agreement with the Sabatier principle for the optimum HER.⁸²

CONCLUSIONS

In conclusion, we have demonstrated a synthetic approach to form efficient and stable bifunctional electrocatalysts in alkaline water electrolysis. Specifically, electrochemical Te dissolution from mechanically exfoliated NiTe₂ followed by mild oxidation results in multidomain NiTe_{2-x}O₈/NiO_x heterostructures, and compressive and tensile strains developed in the materials by surface buckling resulted in the alkaline OER/HER exceeding that of IrO_x/Pt on centimeter-scale electrodes. MLIP and DFT calculations evaluated the atomic-scale active sites from the possible domain models composing the heterostructures, thus revealing the OER/HER enhancement mechanisms associated with the spillover of intermediate species among adjacent domain surfaces, which provides reduced energy barriers in the corresponding multistep reactions. Since all the processes in this work are scalable with reliable material transfer techniques,

we anticipate that use of our heterostructured catalysts as core components will be integrated into a range of electrochemical energy devices, such as proton- and anion-exchange membrane electrolyzers and alkaline air batteries. While the present material system is not yet optimized for cost, the overall fabrication cost can be significantly lowered by adopting inexpensive electrode substrates without sacrificing performance. Taken together, our results demonstrate a robust and scalable strategy for developing heterostructured bifunctional catalysts applicable to next-generation electrochemical energy devices.

METHODS

Mechanical Exfoliation and Dry Transfer Processes of Layered NiTe₂. First, a bulk NiTe₂ crystal (CAS: 12035–59–5, supplier: 2d semiconductor) was placed on top of Nitto thermal release tape (or base tape). In detail, approximately 0.2 mg of NiTe₂ crystals was placed at the center of segmented (1 cm × 1 cm) regions on the base tape, repeatedly cleaved using a carrier tape (i.e., a second Nitto thermal release tape with a similar size), and then stepwise retransferred onto the base tape (Figure S1). In this procedure, the carrier tape was also divided into segmented regions and carefully aligned with the base tape during each step for a single cycle. As the number of exfoliation/transfer cycles increased, progressively more uniform lateral distributions of NiTe₂ flakes were obtained across the segmented regions of the base tape. Optical images at different cycle numbers (1, 10, 50, and 100) clearly demonstrate the reproducibility of the process and the systematic increase in flake coverage and uniformity (Figure S2). To prepare the Au/PS substrates, biaxially prestrained, thermoplastic PS films (Grafix shrink film) were rinsed with ethanol and distilled water following the deposition of Au films (thickness $\sim 25\text{ nm}$) using thermal evaporation. After forming the adhesion between the substrate and the tape, the resulting sample was heated and pressed with a weight (500 g for the sample area of 5 cm × 5 cm) at 110 °C for 5 min on a hot plate, and then the tape was removed.

Electrochemical Te Dissolution from Layered NiTe₂ And Subsequent Oxidation to Form NiTe_{2-x}O₈/NiO_x Heterostructures. We electrochemically formed Te vacancies on the mechanically exfoliated NiTe₂ within the conventional three-electrode cell using cyclic voltammetry (CV). Here, we first prepared the working electrode (WE) based on the NiTe₂/Au/PS; we attached copper (Cu) wires to the local areas of the exposed Au films (not covered by NiTe₂ flakes) by using electrically conductive silver (Ag) paste (Silver paste P-100, CANS). To protect the electrical contact areas on the samples from the electrolyte, we covered the Cu wires with Teflon tubes following the passivation of the contact regions using epoxy resin adhesive (CEMEDINE). Before the electrochemical Te dissolution from the NiTe₂/Au/PS working electrode (WE), dry nitrogen (N₂) was passed through the 1 M KOH aqueous solution with a 1 L/min flow rate for 5 min. We applied reductive CV with scan rate of 100 mV/s between 0 V to −0.9 V vs RHE for 30 cycles using a ZIVE SP1 potentiostat (WonATech). The CV conditions were selected to dissolve Te atoms from NiTe₂ flakes as reported from previous literatures.^{37,38} During the electrochemical processes, dry N₂ was bubbled in the electrolyte with a 1 L/min flow rate. After Te dissolution, we took the WE out from the cell and quickly oxidized it in a convection oven for 30 min at 60 °C. After oxidation, the samples (the electrical contact areas were cut away from the sample) were stored under atmospheric conditions for 1 day before the following strain-engineering processes, surface measurements, and characterization. We verified the sample-to-sample consistency of electrocatalyst surface coverage on electrode substrates using MATLAB image analysis (Figure S3). Furthermore, the average size and thickness of the heterostructure flakes in each sample were confirmed to be highly uniform, as statistically validated by atomic force microscopy (Table S1).

Strain Engineering of NiTe_{2-x}O₈/NiO_x Heterostructures to Fabricate the Working Electrodes for OER/HER Measure-

ments. For the formation of buckled $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ flakes along with Au wrinkle formation, the samples (i.e., the heterostructures on Au/PS substrates) were loaded in a convection oven at 125 °C to relieve prestrain on the PS substrate, because we used a biaxially prestrained PS film (Grafix shrink film, Shrink Film, <https://www.grafixarts.com/products/shrink-film/>) which can be shrunk to ~20% of its original (i.e., prestrained) size by heating the film above its glass transition temperature of 125 °C. This behavior means that we are able to relieve the prestrain (uniaxially or biaxially) by up to $\varepsilon \sim 0.8$.^{22,83,84} To induce uniaxial compressive strains, we mechanically clamped each end of the PS substrates before heating above the glass transition temperature of PS. The PS substrate was heated until the target strain was reached and then cooled to room temperature to stop shrinking. As a result of the strain relief, the PS substrate shrank, and compressive strain was induced in the $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ flakes and the Au film, forming surface wrinkles. Macroscopic strain (ε) was determined from the change in distance between two lines drawn on the sample before and after the shrinking process using the equation $\varepsilon = (l_0 - l_t)/l_0$, in which l_t is the final length and l_0 is the initial length.⁴⁶ To prepare the WEs for OER/HER using the strain-engineered $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$, we iterated the aforementioned fabrication processes; the copper (Cu) wires were attached to the surfaces of the local areas of Au wrinkles without buckled $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ using Ag paste. To protect the electrical contact areas on the samples from the electrolyte, we again covered the Cu wires with Teflon tubes following the passivation of the contact regions using epoxy resin adhesive (Figure S6).

OER/HER Measurements Using Three-Electrode Cells Containing $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ Heterostructures. 1.0 M KOH aqueous solution electrolyte was prepared by mixing 14.0 mL of 5 M KOH (Daejung Chemicals) and distilled water until the total mixture volume was 70.0 mL. We constructed a three-electrode cell configuration in a three-neck flask, using flame-cleaned Pt foil as a counter electrode, Hg/HgO as a reference electrode (+0.926 V vs RHE), and the $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ heterostructured catalyst (before and after straining) fabricated in this study as the WEs. We characterized the OER/HER performances of the electrochemical cells using a ZIVE SP1 potentiostat (WonATech). Dry nitrogen (N_2) was bubbled in the electrolyte for 5 min before the electrochemical measurements. During the OER/HER measurements, dry N_2 was passed through the electrolyte with a 1 L/min flow rate, and the dynamic iR correction feature was turned on to monitor and autocorrect the potential-dependent iR drop on a real-time basis. Independent from the geometric area, the C_{DL} measured by CV was used to normalize the current to account for active sites. For the linear sweep voltammetry (LSV) over different overpotential ranges, we set a constant scan rate of 0.2 mV/s.

Calibration of Hg/HgO Reference Electrode. The calibration of the Hg/HgO electrode was performed in a standard three-electrode system. To construct the electrochemical cell for the calibration, Pt foils were used as the working and counter electrodes, and the Hg/HgO electrodes were used as the reference electrodes. Electrolytes were prepurged and saturated with high-purity H_2 . CV was run at a scan rate of 1 mV s^{-1} , and the average potential with j of zero mA/ cm^2 during a single cycle of CV was considered to be the thermodynamic potential for the hydrogen electrode reaction referring to the reported process.⁸⁵ The Hg/HgO reference electrode used during the OER/HER stability tests was carefully monitored by referring to the calibration CV curves, ensuring that potential drift in the reference electrode was minimized (Figure S47).

Electrochemically Active Surface Area (ECSA) Estimation by Cyclic Voltammetry (CV). To characterize the electrochemically active surface area (ECSA), the 477 electrochemical double-layer capacitance (C_{DL}) of the catalytic surfaces in each three-electrode cell (containing $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ heterostructures strained at different ε) was measured. We characterized C_{DL} , which is linearly proportional to ECSA, by measuring the non-Faradaic capacitance current from the scan-rate dependence of CV (Figure S21). Here, we assumed that all the capacitance current is due to double-layer charging as reported in previous literature.⁵⁶ To measure double-layer charging via CV, a

potential range in which no apparent Faradaic process occurs was determined from static CV, for which the range in this work was centered at the open-circuit potential (OCP) of the system. The C_{DL} was estimated by plotting Δj , the current density difference between the charging and discharging processes at OCP, as a function of the scan rate from 20 to 100 mV/s, where the slope is twice the value of C_{DL} .⁵⁵

Electrochemical Full-Cell Measurements Using a Lab Scale, Two-Electrode Electrolyzer. A two-electrode setup was used to demonstrate the electrochemical full-cell measurements; the ($\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$)/Au/PS samples were used as a cathode as well as an anode (Figure S33). We used a conventional three-neck flask configuration to construct the electrolyzer; 70.0 mL of aqueous electrolyte was filled in a cell. Before measurements, electrolytes were purged with dry nitrogen (N_2) for 5 min to eliminate unknown gas species. A ZIVE SP1 potentiostat (WonATech) was used to take the two-electrode full-cell measurements with a scan rate of 5 mV/s.

Surface Topography Measurements Using Atomic Force Microscopy (AFM). We statistically analyzed the thickness and lateral size distributions of as-transferred NiTe_2 and $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ flakes using AFM (MultiMode 8-HR, BRUKER). All the measurements were performed in tapping mode using silicon nitride cantilevers with silicon tips (NCHV-A, BRUKER; resonance frequency of 320 kHz and nominal tip radius of <10 nm).

Transmission Electron Microscopy (TEM) to Characterize NiTe_2 and $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$. We performed TEM analyses to characterize the crystal structures of the mechanically exfoliated NiTe_2 and $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ heterostructures. First, poly(methyl methacrylate) (PMMA) was spin-coated on Au/SiO₂/Si substrates with transferred NiTe_2 and processed $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ flakes at 1,500 rpm for 1 min, followed by baking on a hot plate at 120 °C for 5 min. We then separated PMMA/ NiTe_2 (and PMMA on $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$) films from the Au/SiO₂/Si substrates by etching the Au films using the gold etchant (TFA, Transene). Upon etching of the Au, the PMMA/ NiTe_2 and PMMA on $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ films were left floating on the etchant solution. Using clean Si substrates, these films were transferred to deionized water several times to rinse off the gold etchant. The resulting films were transferred onto the Quantifoil Cu 1.2/1.3 300 mesh grid (658–300-Cu, Ted Pella) and then baked on a hot plate at 130 °C for 1 h. After baking, the grids were soaked in acetone at 60 °C for 60 min to remove PMMA. The grids were then rinsed with acetone and isopropyl alcohol (IPA) sequentially and then dried on filter paper for 30 min.⁸⁶ High-resolution TEM (HRTEM) analysis was carried out using a TEM (TITAN 80–300, ThermoFisher, US) operated at 300 kV. The available point resolution was better than 1 Å at this accelerating voltage. Images of the samples were recorded by a 4k × 4k CCD (Oneview 109S, Gatan, US) camera. The chemical elemental analyses were carried out using a super-X EDS system in a TEM (TALOS F200X, ThermoFisher, US) operated at 200 kV, and the available point resolution was better than 1.6 Å at this accelerating voltage. Again, the images of the samples were recorded by a 4k × 4k CCD (Ceta 16M, ThermoFisher, US) camera.

X-ray Photoelectron Spectroscopy (XPS) to Characterize NiTe_2 and $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ Before and After Strain-Engineering. XPS was performed in a K-Alpha XPS system (Thermo Scientific) with an Al K α (1487 eV) X-ray source calibrated using the C 1s peak with 400 μm beam spot diameter covering multiple NiTe_2 and $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ flakes. XPS was used to observe the Ni 2p, Te 3d and O 1s peaks of the materials system with spectroscopic variations. The XPS valence band spectra were corrected by subtracting the Shirley background. For accurate comparisons of all valence band spectra, the d-band centers of gravity (E_d) of the samples were calculated in the range of –10.0 to 0 eV, using the equation below, where $N(E)$ is the DOS, E_d is the center of the d-band, and E is the binding energy:^{87,88}

$$E_d = \frac{\int N(E)E dE}{\int N(E) dE}$$

Ultraviolet Photoelectron Spectroscopy (UPS) to Characterize $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ Heterostructures Before and After Strain Engineering. UPS was performed in a Nexsa UPS system (Thermo Fisher Scientific) with a He I (21.22 eV) source with a 1 mm beam spot diameter at a base pressure of 2.0×10^{-8} Torr. For UPS measurements, the samples were biased at -10.00 V to measure the work function, and the relative valence band maximum of $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ heterostructures before and after straining was observed at 0 V. A clean gold sample was used to calibrate the Fermi edge of the spectrometer. To prepare the samples with strained $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$, aqueous PVA (poly(vinyl alcohol), DAEJUNG chemical) solution was first spin-coated on the strained heterostructures (on wrinkled Au/PS substrates) at 2500 rpm for 1 min. Here, aqueous PVA solution was prepared by dissolving 300 mg of PVA powder in 40 mL deionized water and stirring at 120°C for 2 h.⁸⁹ The samples were then baked on a hot plate at 100°C for 2 min. Next, PMMA was spin-coated on the PVA/ $(\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x)/\text{Au/PS}$ at 2,500 rpm for 1 min and then baked on a hot plate at 100°C for 5 min. The samples were then floated on the surface of gold etchant (TFA, Transene) at 0°C to etch away the gold film. Upon etching of the gold film, resulting PMMA/PVA/ $(\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x)$ films were left floating on the etchant solution. Using clean Si substrates, the films were transferred on the surface of methanol to rinse off the gold etchant. Here, we used the ITO-coated substrate to scoop out the films following the baking on a hot plate (at 130°C for 60 min). After the treatments, the samples were soaked in deionized water at 130°C for 120 min to remove PMMA/PVA. Finally, the samples were rinsed with acetone and IPA and dried on filter paper.

Raman Measurements to Characterize NiTe_2 , $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ Before and After Strain Engineering. Raman spectra of the NiTe_2 and $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ flakes on Au/PS substrates were obtained using Raman microscopes (SENTERRA II, BRUKER, UniRAM, UniNanoTech). The measurements were performed in the back-scattering configuration with 532 nm excitation from a diode-pumped solid-state laser. The excitation was focused using a $100\times$ microscope objective with a numerical aperture of 0.9, resulting in a spot diameter of $\sim 1\ \mu\text{m}$, capable of probing single flakes. The scattered signal was sent through an 1800 gr/mm grating. For UniRAM Raman microscope, the scattered signal was collected using a thermoelectrically cooled charge-coupled device (CCD) detector (iDUS DV401A, Andor SS Technology). Spectra were acquired under low laser power (1.5 mW) to prevent sample damage. Typical acquisition times were 4 s per spectrum. All measurements were performed under ambient conditions. Raman mapping was conducted using a motorized stage in the UniRAM Raman microscope with a step size of 300 nm in both the x and y directions. Raman images were constructed using specific parameters (position/intensity/shift of A_{1g} and E_g modes) based on the Raman spectra recorded at each point.

In-Situ Raman Measurements to Characterize Real Time Water Electrolysis. The in situ Raman spectroscopy was carried out in 1.0 M KOH electrolyte, which was prepared by mixing 200 mL of 5 N KOH (Daejung Chemicals) and distilled water until the total mixture volume reached 1.0 L. We constructed a three-electrode cell configuration in a Raman electrochemical flow cell (Raman EQCM FC, Redoxme AB), using strain-engineered $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ catalysts fabricated in this study as the working electrode (WE), and Pt wire (Redoxme AB) as a counter electrode (CE) and 1 M KOH-filled Hg/HgO electrode as a reference electrode (RE, -0.926 V vs RHE, WizMAC). Spectra were acquired using a Raman microscope (SENTERRA II, BRUKER) while the electrode potential was controlled by using a ZIVE SP1 potentiostat (WonATech). The electrode potential was sequentially applied from open-circuit potential (OCP) to -0.25 V vs RHE in 0.05 V increments for HER and from OCP to 0.80 V vs RHE in 0.20 V increments for OER, with each step maintained for 60 min, during which Raman spectroscopy was performed. Flow rate of 2 mL/min was maintained using a peristaltic pump (MODEL A, Redoxme) for all measurements.

Measurements of Water Contact Angle to Characterize the Surface Wetting Properties of Samples. To characterize the

surface wettability of $(\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x)\text{Au/PS}$ surfaces, we measured the static water contact angle (θ_w), advancing contact angle (θ_A), and receding contact angle (θ_R) at room temperature using a surface analyzer (DROP SHAPE ANALYZER – DSA25, KRÜSS, DE). For every measurement, we used an initial water droplet size of $6\ \mu\text{L}$, and at least ten measurements for each sample were averaged. To calculate contact angle hysteresis (θ_H), the measured θ_R was subtracted from the measured θ_A .⁹⁰

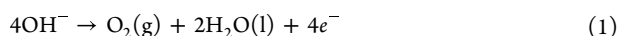
Measurement of the Gaseous Product in Real Time to Calculate Faradaic Efficiency. To measure the Faradaic efficiency, an Agilent GC 8890A with a thermal conductivity detector was used. The GC system was calibrated using a certified standard of hydrogen and oxygen before use (Shinyang Medicine Co. Ltd., Korea). GC measurements were conducted by connecting an airtight electrochemical cell—equipped with carrier-gas inlet and outlet lines—to the gas chromatograph. The injection gas flow rate was controlled at 5 mL/min using a flow meter connected to the argon carrier-gas line and through the front inlet control of the GC. To ensure complete sealing of the electrochemical cell, the gas inlet and outlet were connected using gas connectors wrapped with Teflon tape, while the working, counter, and reference electrodes were fixed with flangeless male nuts and O-rings to prevent air leakage. The Faradaic efficiency measurements were performed using an airtight electrochemical cell purged with Ar for 30 min to remove any residual gases, thereby minimizing the influence of impurities, and then started the measurements under sealed conditions. We used the method from the published literature to calculate the theoretical amount of O_2 and H_2 for the given experimental conditions.⁹¹

MLIP and DFT Calculations. MLIP calculations were employed to scan and evaluate the active sites on NiTe_2 - and NiO-based surface models for the electrochemical intermediate adsorptions during alkaline OER and HER. The Preferred Potential (PFP) implemented in the Matlantis platform (Preferred Computational Chemistry, Inc.) was used with a dispersion-corrected configuration to describe interatomic interactions.^{92–94} Geometry optimizations were performed using the limited-memory BFGS algorithm in the Atomic Simulation Environment (ASE) with a force convergence criterion of $0.001\ \text{eV}\ \text{\AA}^{-1}$.⁹⁵ Transition state calculations were carried out using the climbing image–nudged elastic band (CI–NEB) method with the fast inertial relaxation engine (FIRE) algorithm and a force threshold of $0.02\ \text{eV}\ \text{\AA}^{-1}$.^{96,97} DFT calculations were performed using the Vienna ab initio simulation package (VASP, version 5.4.4), with the Perdew–Burke–Ernzerhof (PBE) exchange–correlation functional and D3 (BJ) dispersion correction.^{98–100} The projector-augmented wave (PAW) method was used with a plane-wave cutoff energy of 450 eV.¹⁰¹ Here, electronic and ionic convergence thresholds were set to 10^{-5} eV and $0.02\ \text{eV}\ \text{\AA}^{-1}$, respectively. Brillouin zone sampling was then performed with a $4 \times 4 \times 1$ Monkhorst–Pack k -point mesh. For NiO-based models, spin-polarized calculations were carried out with a Hubbard U correction ($U - J = 5.3$ eV) applied to Ni 3d orbitals.¹⁰² In the literature where sensitivity analyses on Hubbard U values for NiO and its surfaces were performed,^{103–105} this value has shown to yield physically reasonable properties such as band gaps, defect energetics, and magnetic moments.^{106–109} Bader charge analyses were performed using the algorithm developed by the Henkelman group.^{110–112}

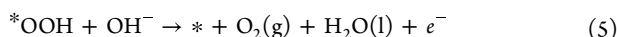
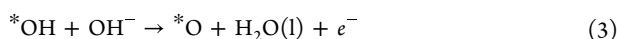
Surface models were constructed based on trigonal NiTe_2 (space group $P\bar{3}m1$) and cubic NiO (space group $Fm\bar{3}m$) systems. The NiTe_2 (001) surface was modeled using $11.37 \times 11.37 \times 15.76\ \text{\AA}^3$ supercell with a vacuum spacing of $15\ \text{\AA}$ (Figure S37). Uniaxial tensile and compressive strains ($\pm 5\%$), calculated from experimental Raman shifts, were applied along the x and y directions to evaluate the effects of mechanical deformation in the electrochemical reactions. The models with vacancies were prepared by removing two equivalent surface Te atoms under the slab symmetry, while the models with O substitutions were constructed by substituting two additional Te atoms with O (Figures S38–S39). The NiO (100) surface was modeled using $8.4 \times 8.4 \times 11.4\ \text{\AA}^3$ supercell with a vacuum region of $15\ \text{\AA}$. A single surface Ni atom was removed to generate the NiO-V model. Counting atoms in the topmost surface unit cell, the NiTe_2

models correspond to $\text{Ni}_9\text{Te}_{18}$ (pristine), $\text{Ni}_9\text{Te}_{16}$ (two surface Te vacancies), and $\text{Ni}_9\text{Te}_{14}\text{O}_2$ (two surface Te vacancies plus two O substitutions), while the NiO models correspond to Ni_8O_8 (pristine) and Ni_7O_8 (one surface Ni vacancy). These local stoichiometries were selected to represent defect-rich edge and surface motifs indicated by XPS, Raman, and TEM-EDX, providing models of the local coordination environments that may govern catalytic reactivity. All models were fully relaxed after applying the directional strains to find thermodynamically stable structures, and then OER/HER reactivities on the deformed models were evaluated.^{22,40,106,113}

Free Energy Calculations for Alkaline OER/HER. The free energy diagrams for alkaline (pH = 14) OER were obtained based on the overall reaction.^{114,115}



This process was modeled via four-step proton-coupled electron transfer processes involving the successive adsorptions of $^*\text{OH}$, $^*\text{O}$, and $^*\text{OOH}$ intermediates.¹¹⁶



The four-step processes were considered based on conventional adsorbate evolution mechanism (AEM), the model for alkaline OER on transition metal surfaces.^{117,118} For each step ($n = 1 - 4$) in the processes, the Gibbs free energy change (ΔG_n) was calculated as follows:¹¹⁹

$$\Delta G_1 = E_{^*\text{OH}} - E_* - E_{\text{H}_2\text{O}} + \frac{1}{2}E_{\text{H}_2} + (\Delta\text{ZPE} - T\Delta S)_1 - eU_{\text{RHE}} \quad (6)$$

$$\Delta G_2 = E_{^*\text{O}} - E_{^*\text{OH}} + \frac{1}{2}E_{\text{H}_2} + (\Delta\text{ZPE} - T\Delta S)_2 - eU_{\text{RHE}} \quad (7)$$

$$\Delta G_3 = E_{^*\text{OOH}} - E_{^*\text{O}} - E_{\text{H}_2\text{O}} + \frac{1}{2}E_{\text{H}_2} + (\Delta\text{ZPE} - T\Delta S)_3 - eU_{\text{RHE}} \quad (8)$$

$$\Delta G_4 = E_* - E_{^*\text{OOH}} + 2E_{\text{H}_2\text{O}(\text{l})} - \frac{1}{2}E_{\text{H}_2} + \Delta G_0 + (\Delta\text{ZPE} - T\Delta S)_4 - eU_{\text{RHE}} \quad (9)$$

Here, E_* and E_{X^*} denote the DFT total energies of the model surface before and after intermediate X adsorption, respectively. $E_{\text{H}_2\text{O}}$ and E_{H_2} are the total energies of corresponding isolated molecules. Thermodynamic corrections (ΔZPE , ΔS) were applied at 298.15 K based on the values from literatures.^{71,120–122} The total free energy for water electrolysis (ΔG_0) on model surfaces set as 4.92 eV, consistent with experimental thermochemistry. The applied potential U was referenced to the reversible hydrogen electrode (RHE), and all the calculations were performed at $U = 0$ V to reflect standard equilibrium conditions. The theoretical per-site OER overpotential (η) was obtained from the largest ΔG_n as

$$\eta = \frac{1}{e} \times \max\{\Delta G_1, \Delta G_2, \Delta G_3, \Delta G_4\} - 1.23 \text{ V} \quad (10)$$

For alkaline HER, the free energy of hydrogen adsorption (ΔG_{H^*}) was evaluated as

$$\Delta G_{\text{H}^*} = \Delta E_{\text{ads}}^{\text{H}} + \Delta E_{\text{ZPE}} + T\Delta S \quad (11)$$

where $\Delta E_{\text{ads}}^{\text{H}} = E_{\text{H}/\text{slab}} - E_{\text{slab}} - \frac{1}{2}E_{\text{H}_2}$. The thermodynamic correction term was taken to be 0.20 eV at 298.15 K, in accordance with previous literature.¹²³

Evaluations of Atomistic OER/HER (Re)activity with Volcano Plots. Theoretical OER activity trends for the atomic sites from supercell models were analyzed by plotting η as a function of the ΔG_2 . This OER volcano plot was constructed using the linear scaling relation reported from previous literatures:^{73,124}

$$\Delta G_{^*\text{OOH}} = \Delta G_{^*\text{OH}} + 3.2\text{eV} \quad (12)$$

which defines a theoretical minimum η of ~ 0.37 V.¹²⁵

Theoretical HER activity trends for the atomic sites were evaluated in HER volcano plot by correlating the calculated exchange current density (j_0) with ΔG_{H^*} .¹²⁶ When $\Delta G_{\text{H}^*} < 0$, j_0 values were calculated as

$$j_0 = -ek_0 \frac{1}{1 + \exp(-\Delta G_{\text{H}^*}/k_{\text{B}}T)} \quad (13)$$

and for the case of $\Delta G_{\text{H}^*} > 0$, j_0 values were calculated as follows:

$$j_0 = -ek_0 \frac{1}{1 + \exp(-\Delta G_{\text{H}^*}/k_{\text{B}}T)} \exp(-\Delta G_{\text{H}^*}/k_{\text{B}}T) \quad (14)$$

where $k_0 = 200 \text{ s}^{-1}\cdot\text{site}^{-1}$, $T = 298.15 \text{ K}$, and k_{B} is the Boltzmann constant.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsnano.5c14702>.

Experimental procedure for the mechanical exfoliation and transfer of NiTe_2 ; progressive exfoliation of NiTe_2 crystals over repeated mechanical exfoliation cycles; reproducibility of $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ formation across samples; characterization of the thickness and size distributions of NiTe_2 and heterostructures; fabrication of strain-engineered $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ heterostructure catalysts and three-electrode cell system for OER and HER; configuration of working electrode with strained $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ heterostructures; tunable Te vacancy generation to form $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$; quantification of Te vacancies in $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ based on XPS core-level spectra; TEM analyses of as-transferred NiTe_2 ; EDX maps of as-transferred NiTe_2 ; TEM analyses of $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ heterostructures having different lateral lengths; XPS survey spectrum; tunable tensile and compressive strains in $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ using substrate buckling; TEM analyses of $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ heterostructures before and after strain-engineering; statistical analyses of $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ heterostructures with compressive and tensile strains; OER/HER on as-transferred NiTe_2 in comparisons with $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ and Au surface; EIS measurements of $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ heterostructures for OER; electrochemical performance of NiTe_2 before and after strain engineering; OER performance of $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ with varying CV cycles; surface coverage of $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ after 30 and 40 CV cycles; estimations of C_{DL} for ECSA normalization; SEM images of IrO_x coated on carbon cloth (IrO_x/CC) after long-term stability test; OER performance comparisons with reported literature; HER performance of $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ with varying CV cycles; EIS measurements of $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ heterostructures for HER; surface wetting properties of $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ heterostructures

before and after strain-engineering; HER performance comparisons with reported literature; water electrolysis from $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ strained at system maximum strain; water electrolysis from strain-engineered $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ before and after stability test; statistical analyses of strains developed in $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ before and after OER/HER stability test; HER/OER stability of Pt/IrO_x ; OER/HER faradaic efficiencies of strained $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ heterostructures; overall alkaline water electrolysis using two-electrode electrolyzer cell based on $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ heterostructures; polarization curves of strain-engineered $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ heterostructures at high current densities; mass activities of $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ before and after strain-engineering in comparisons with noble-metal catalysts; water electrolysis from strained $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ in different electrolyte concentrations; domain models based on stoichiometric NiTe_2 ; domain models based on $\text{NiTe}_{2-\delta}$; domain models based on $\text{NiTe}_{2-\delta}\text{O}_\delta$; Calculated η values for OER by MLIP and DFT; in situ Raman spectra during OER from $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$; comparisons of η values for OER between single-site and dual-site mechanisms; charge density difference analyses for OER intermediates using single-site and dual-site mechanisms; comparisons of Δq values for OER intermediates; OER comparisons on NiO-V model by adsorbate evolution and lattice oxygen mechanism; experimental evidence for hydrogen spillover in alkaline HER; cyclic voltammetry curves for Hg/HgO reference electrode calibrations; statistical analyses of $\text{NiTe}_{2-\delta}\text{O}_\delta/\text{NiO}_x$ heterostructures across samples (Table S1); OER performance comparisons with reported literature (Table S2); HER performance comparisons with reported literature (Table S3); electrocatalytic performance comparisons with industrial catalysts (Table S4); comparisons of mass activities among reported state-of-the-art catalysts for OER (Table S5); comparisons of mass activities among reported state-of-the-art catalysts for HER (Table S6); MLIP evaluations of OER for NiTe_2 based models (Table S7); MLIP evaluations of OER for $\text{NiTe}_{2-\delta}$ based models (Table S8); MLIP evaluations of OER for $\text{NiTe}_{2-\delta}\text{O}_\delta$ based models (Table S9); Bader charge analyses for OER intermediates (Table S10); MLIP evaluations of OER for NiO and NiO-V models (Table S11) (PDF)

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Author Contributions

○S.S., I.C., and S.K.H. contributed equally to this work. W.K.L. developed the concept and designed the experiments. S.S. fabricated samples and prepared the electrochemical cells to test water splitting performance. S.S. and S.K.H. characterized the samples using AFM and SEM. S.S., H.J. and I.S.K. measured UPS and Raman spectroscopy. S.S., J.H., J.L., and I.S.K. measured XPS spectroscopy. S.S., I.S.K., and M.K.C. characterized the samples using TEM. I.C. and B.S. developed the theoretical models. S.S., D.J.P., I.S.K., and W.K.L. analyzed experimental and theoretical data. W.K.L. supervised the experiments. S.S., I.C., D.J.P., I.S.K., and W.K.L. wrote the manuscript.

Notes

The authors declare no competing financial interest.

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